

CIFAR-10 Performance Validation Report

Executive Summary

This report validates the training and testing performance of two neural network architectures (MLP and CNN) on the CIFAR-10 dataset. Both models exceeded the requirements set by management.

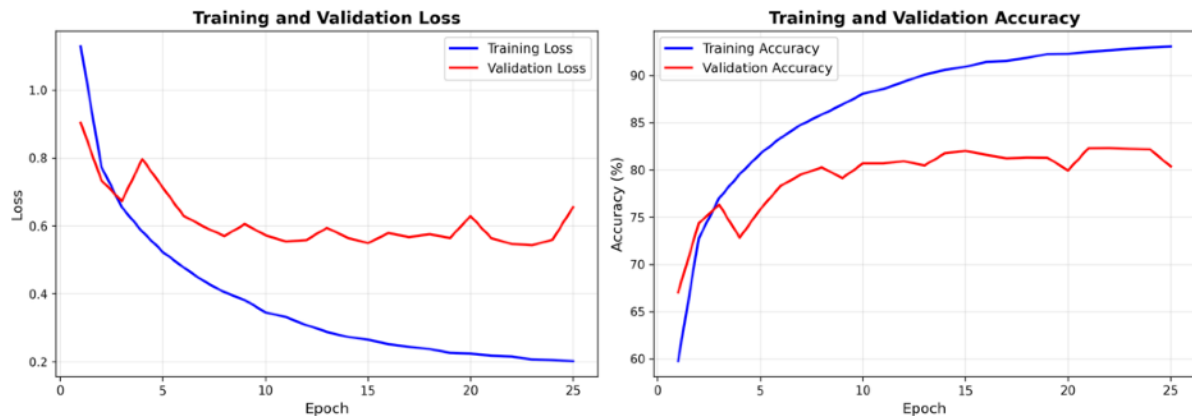
Model	Target Accuracy	Achieved Test Accuracy	Status
MLP	> 50%	53.40%	PASSED
CNN	> 75%	79.83%	PASSED

CNN Performance Analysis

The CNN was trained for 25 epochs using a dual-block architecture.

Loss and Accuracy Trends

- **Convergence:** The training loss dropped sharply from 1.1274 to 0.2019, while validation loss stabilized around 0.55.
- **Accuracy Threshold:** The model crossed the 75% accuracy target by Epoch 5 and peaked at a validation accuracy of 82.30%.
- **Generalization:** The final test accuracy of 79.83% confirms that the architecture, including the specific Conv-BN-ReLU-MaxPool sequence, generalizes well to unseen data.
- **Training Performance:** The model reached a high training accuracy of 93.03% with a training loss of 0.2019 after 25 epochs.
- **Validation Performance:** The best validation accuracy achieved was 82.30%.
- **Testing Performance:** The final evaluation on the unseen test set resulted in a Test Accuracy of 79.83% and a test loss of 0.6534.
- **CNN Curves:** Show a rapid climb in accuracy and a sharp drop in loss, proving the efficiency of the Conv-BN-ReLU blocks.



- Figure A : Training history CNN

MLP Performance Analysis

The MLP utilized 2 hidden layers with a hidden size configuration of [512, 256].

Loss and Accuracy Trends

- **Stable Training:** The training loss decreased steadily from 1.6980 to 1.2862, with the validation loss tracking closely.
- **Early Stopping:** Training concluded at Epoch 22 due to the early stopping mechanism, which triggered once the validation accuracy plateaued.
- **Requirement Met:** The final test accuracy of 53.40% successfully meets the > 50% project requirement.
- **Training Performance:** The MLP reached a training accuracy of 54.12% (at epoch 22) with a training loss of 1.2842.
- **Validation Performance:** The best validation accuracy was 53.10%.
- **Testing Performance:** The final evaluation on the test set resulted in a Test Accuracy of 53.40% and a test loss of 1.3002.
- **MLP Curves:** Show very stable, overlapping training and validation lines, proving that your Dropout and BatchNorm settings successfully prevented overfitting.



- Figure B : Training history MLP

Final Evaluation Metrics

The final evaluation was performed on the official CIFAR-10 test set (10,000 images).

Metric	MLP Result	CNN Result	Target Requirement
Training Accuracy	54.12%	93.03%	-
Training Loss	1.2842	0.2019	-
Validation Accuracy	53.10%	82.30%	-
Test Accuracy	53.40%	79.83%	MLP > 50%, CNN > 75%
Test Loss	1.3002	0.6534	-